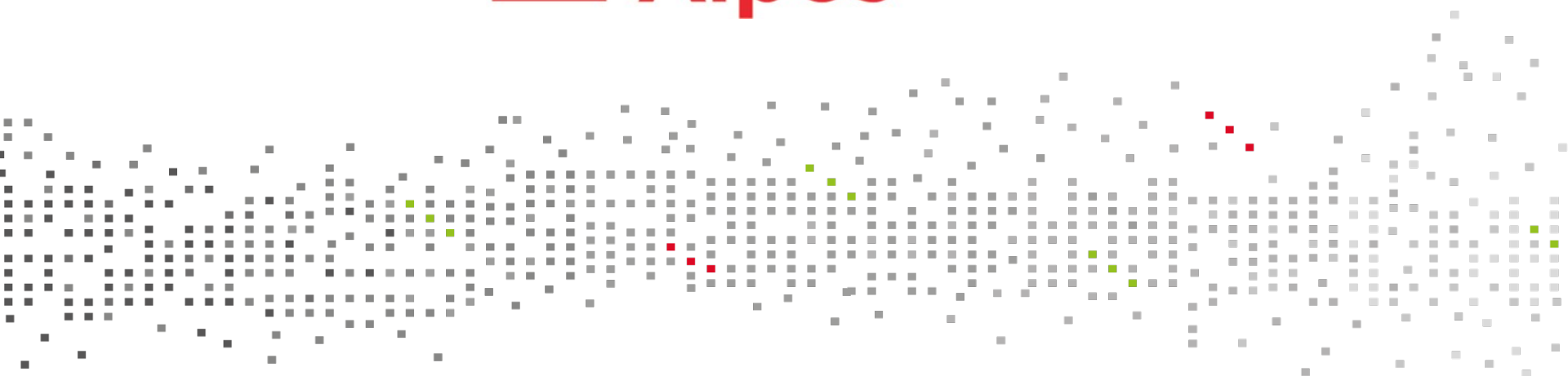




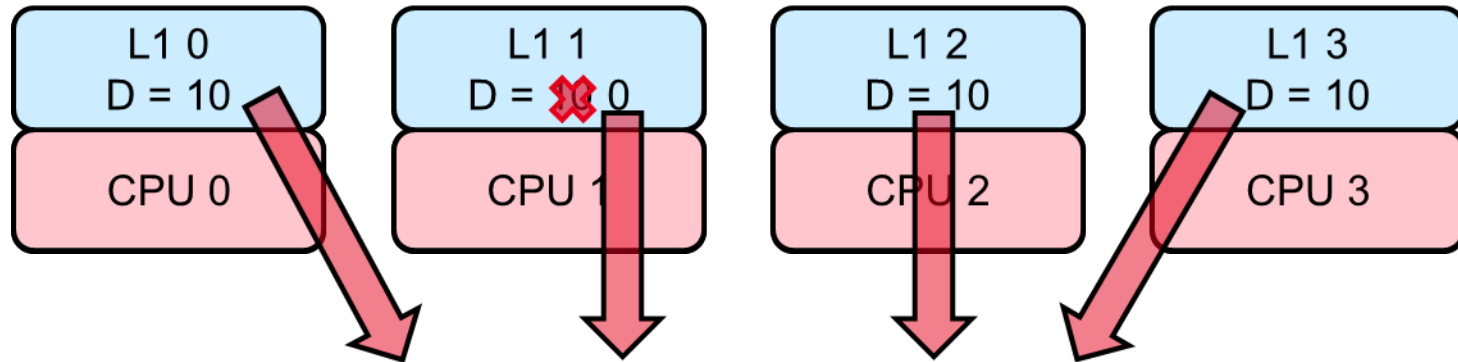
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Alpes**



DYNAMIC COHERENT CLUSTER: A SCALABLE SHARING SET MANAGEMENT APPROACH

ASAP 2018 | **Julie Dumas**, Eric Guthmuller and Frédéric Pétrot | 07/12/2018

- Manycores architecture



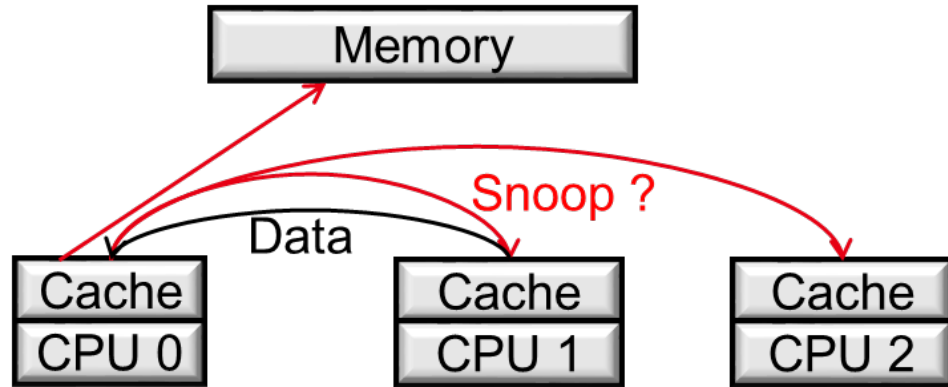
D has not same value in all caches

⇒ Need coherence message to notify that D has a new value

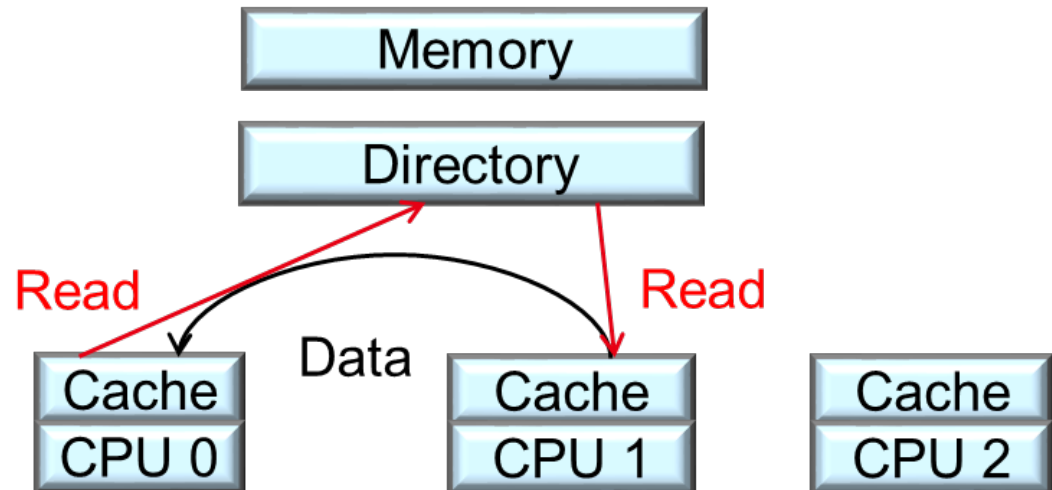
➤ Cache coherence

- Snoop**

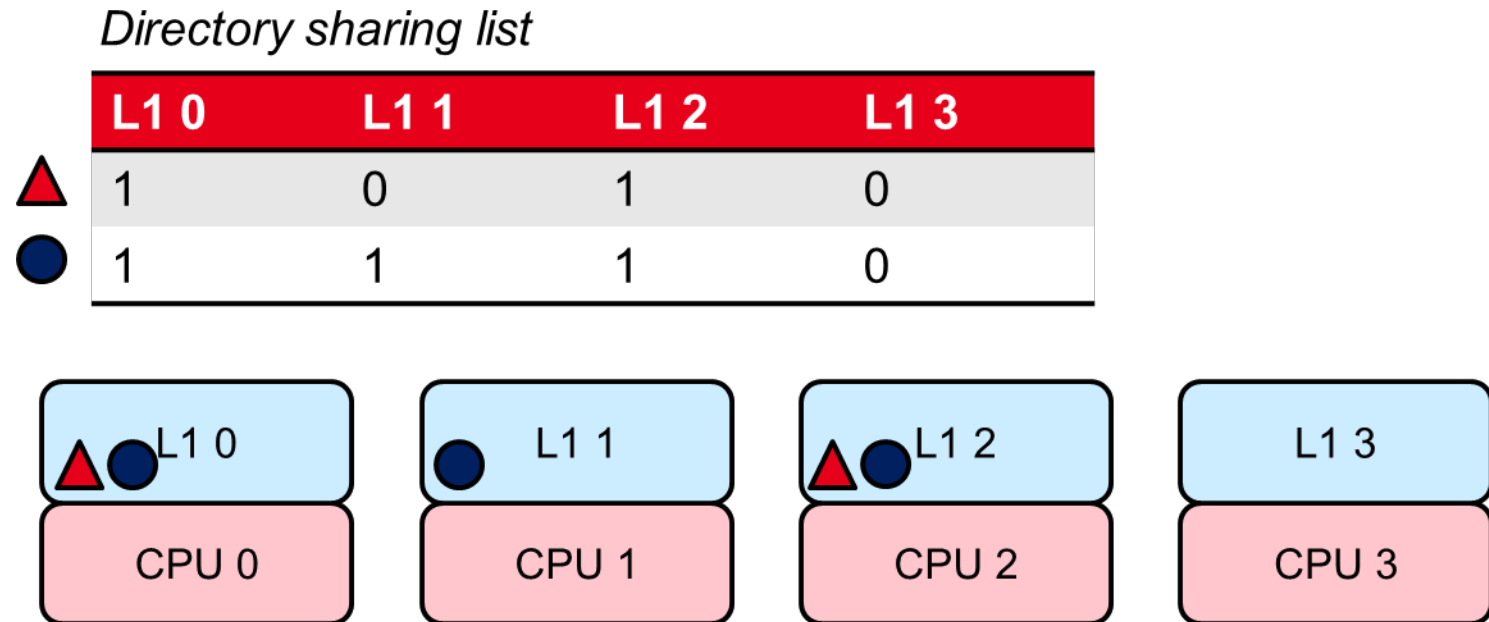
- Based on broadcast
- Problem with NoC



- Directory based**

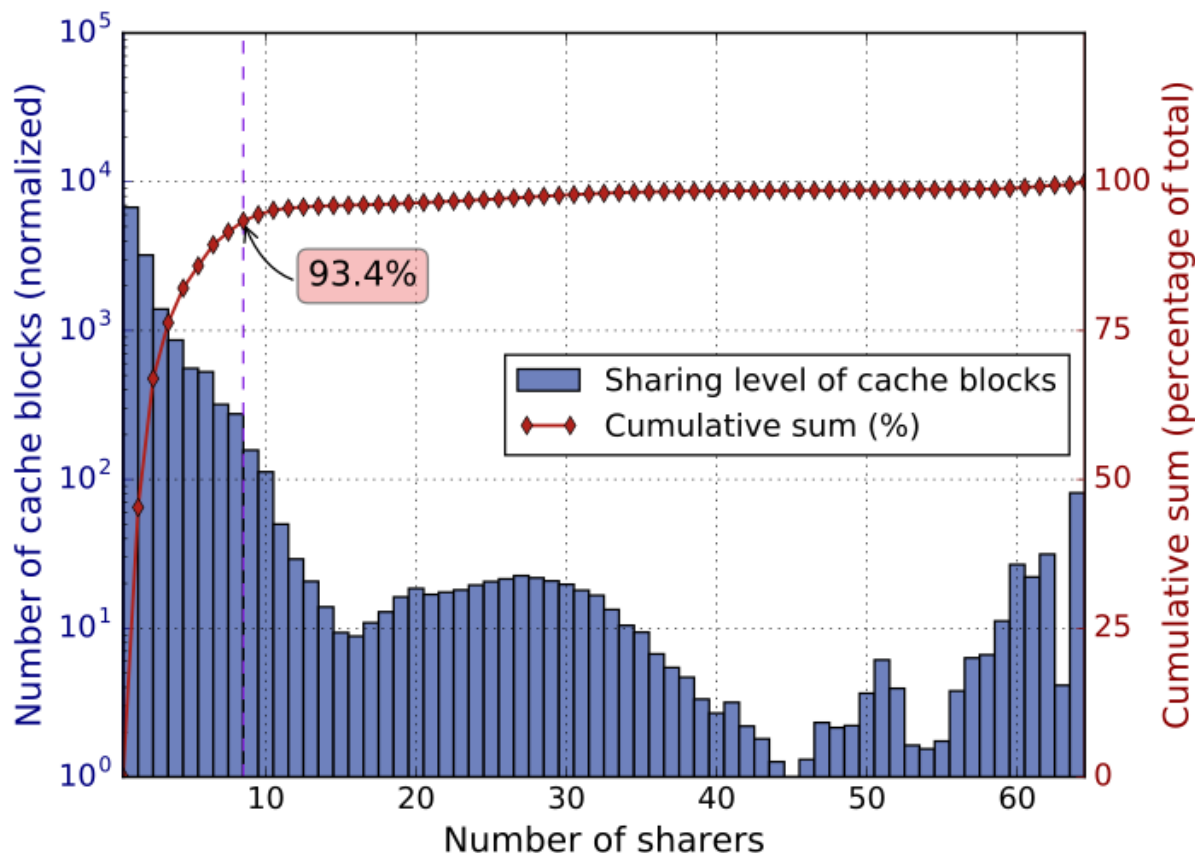


- Full bit-vector



- Problem to scale

- 93.4% blocks are shared by at most 8 cores
 - Benchmark from Parsec and Splash



- **Ackwise [1]**
 - Accurate mode
 - k copies

Global = 0	Keeper	Sharer 2	...	Sharer K
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- Broadcast mode: $\text{nb_copies} > k$
 - Broadcast
 - Sharer count

Global = 1	x	x	x	Sharer Count
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[1] :Kurian *et al.* ATAC, PACT 2010

- **Linked list [2]**
 - Accurate mode :
 - Linked list of sharer
 - Broadcast mode: $\text{nb_copies} > \text{threshold (T)}$ or heap full
 - Broadcast

Directory entry

State	Mode	Sharer	Next
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Heap

Sharer	Next
Sharer	Next
Sharer	Next

[2] :Thapar *et al.* Linked List cache coherence for scalable shared memory, MASCOTS 1993

- **Observations:**

- 94% of cache blocks shared by at most 8 cores
- Sharing level does not increase with the number of cores [3]
- Operating systems map sharers close to other sharers[4]

- **Proposal :**

- Find a region of sharing => rectangle
- Encode sharer in this region with a bit-vector

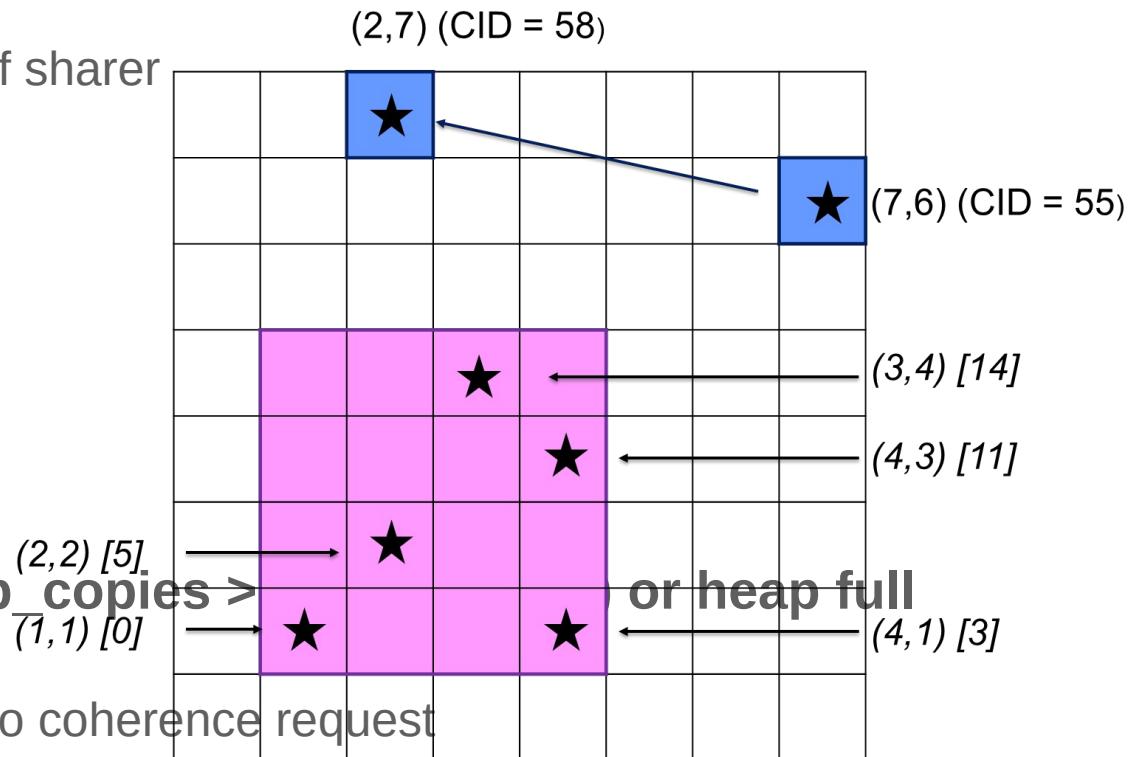
➤ Dynamic Coherent Cluster (DCC)

[3] : Fu *et al.* Coherence domain restriction on large scale systems, MICRO 2015

[4] : Diener *et al.* Affinity-based thread and data mapping in shared memory systems,
Computing Surveys 2016

- Accurate mode:

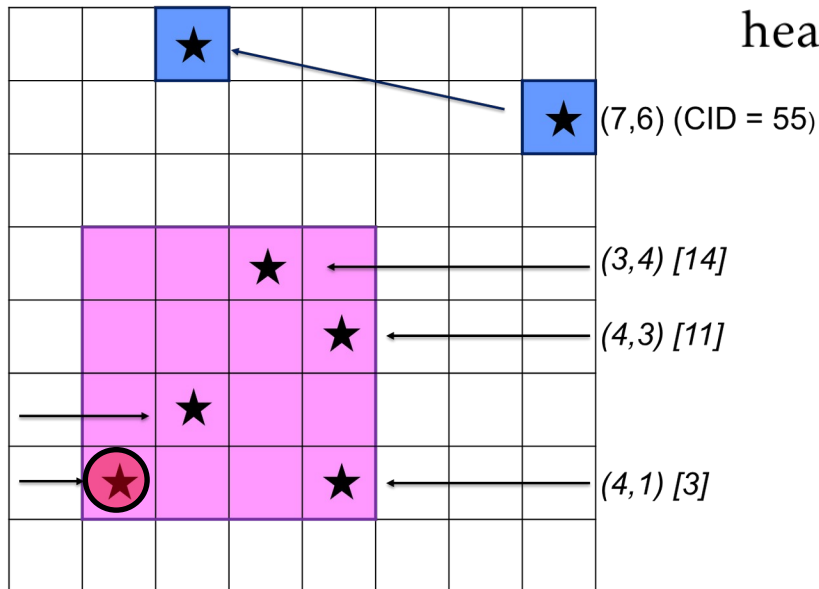
- Rectangle area = C
- Origin and shape evolved during execution
- Outlier: linked list of sharer
 - Initialization
 - Migration



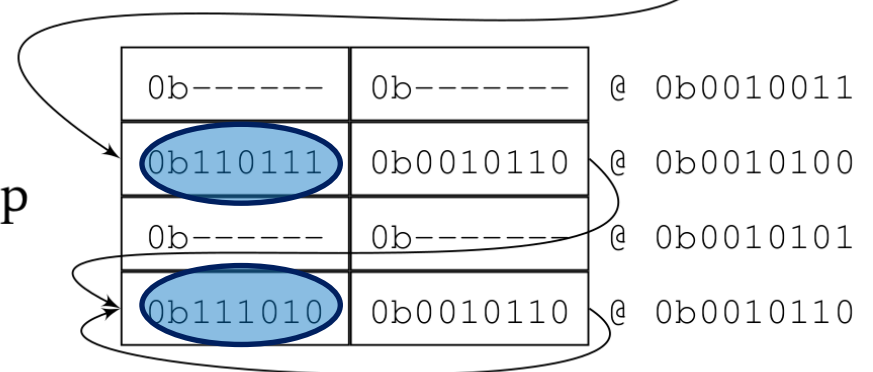
- Broadcast mode:
 - Broadcast
 - Keeper: response to coherence request

broadcast	origin	shape	bit vector	pointer
0b0	0b001001	0b010	0b1001010000010010	0b0010100

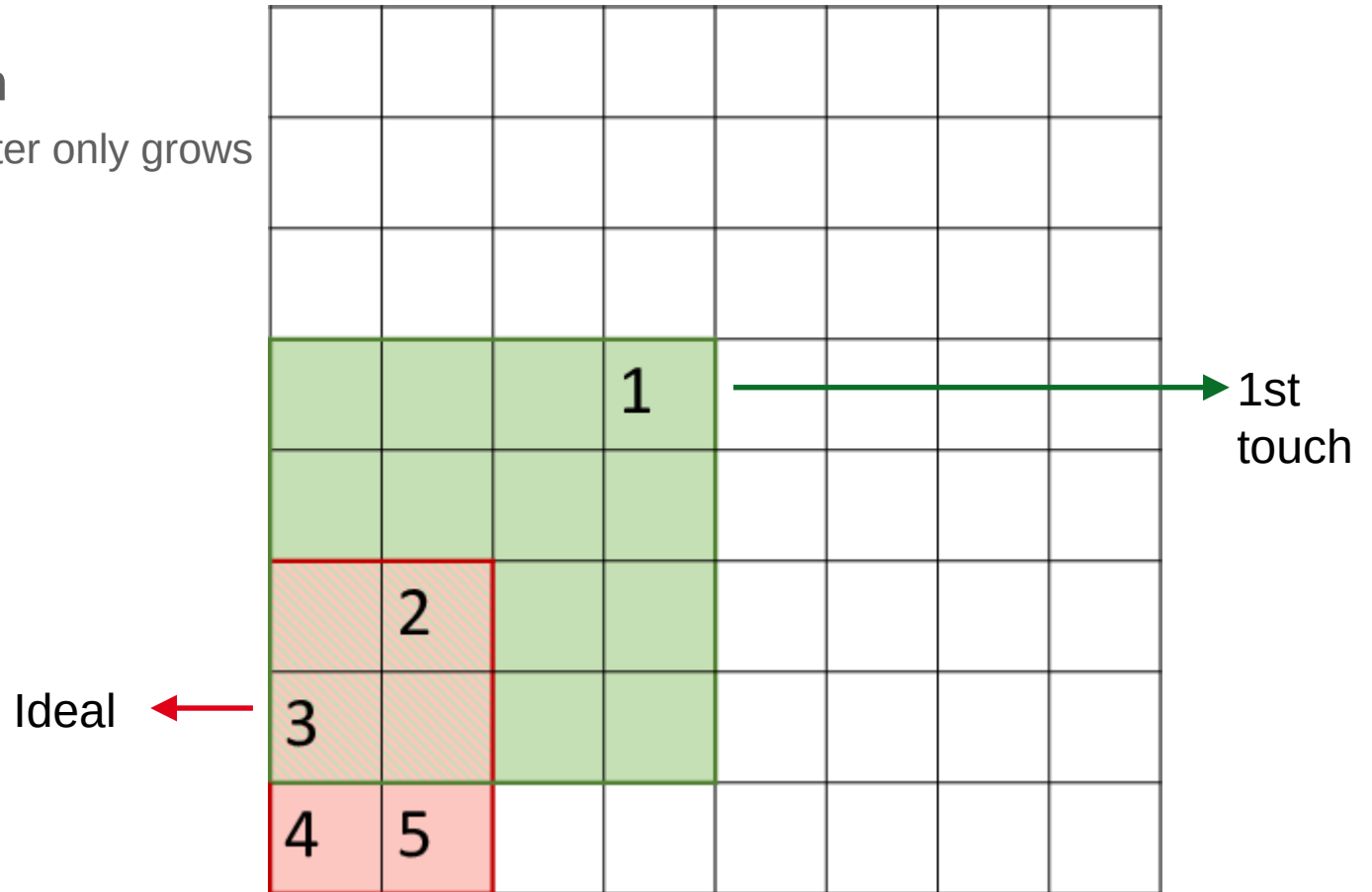
(2,7) (CID = 58)



heap

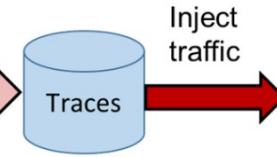
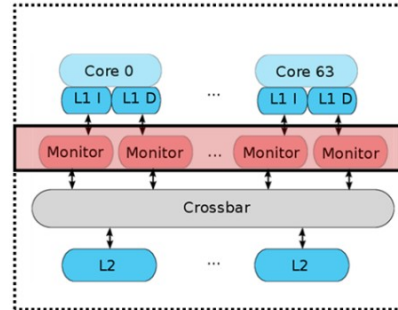


- **Ideal :**
 - Select best cluster
- **First touch**
 - Cluster only grows

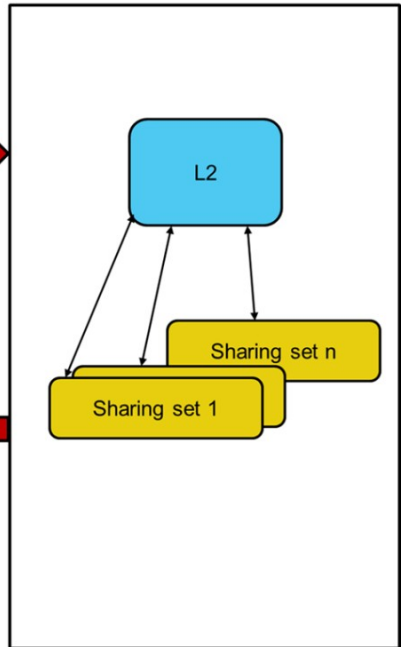


- PyCCExplorer [5]
- 13 applications from Splash2 and Parsec
 - Simmedium
 - Region of Interest (ROI)
- gem5 parameters

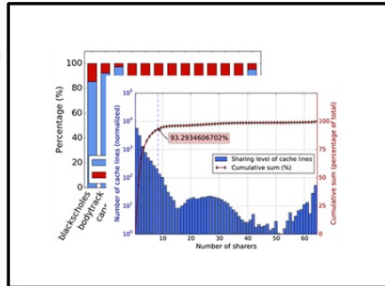
Step 1:
Simulation with gem5
and extract traffic (1 time)



Step 2:
Simulation with a high
level cache model



Step 3:
Analyze



CPU	64 cores Alpha 2GHz
L1 instructions	32kB, 2 way set associative
L1 data	64kB, 2 way set associative
Shared L2	256kB x 64, 8 way set associative
Block size	64B

- Cluster size:

- 4-8-16 => 16

- Linked list threshold T:

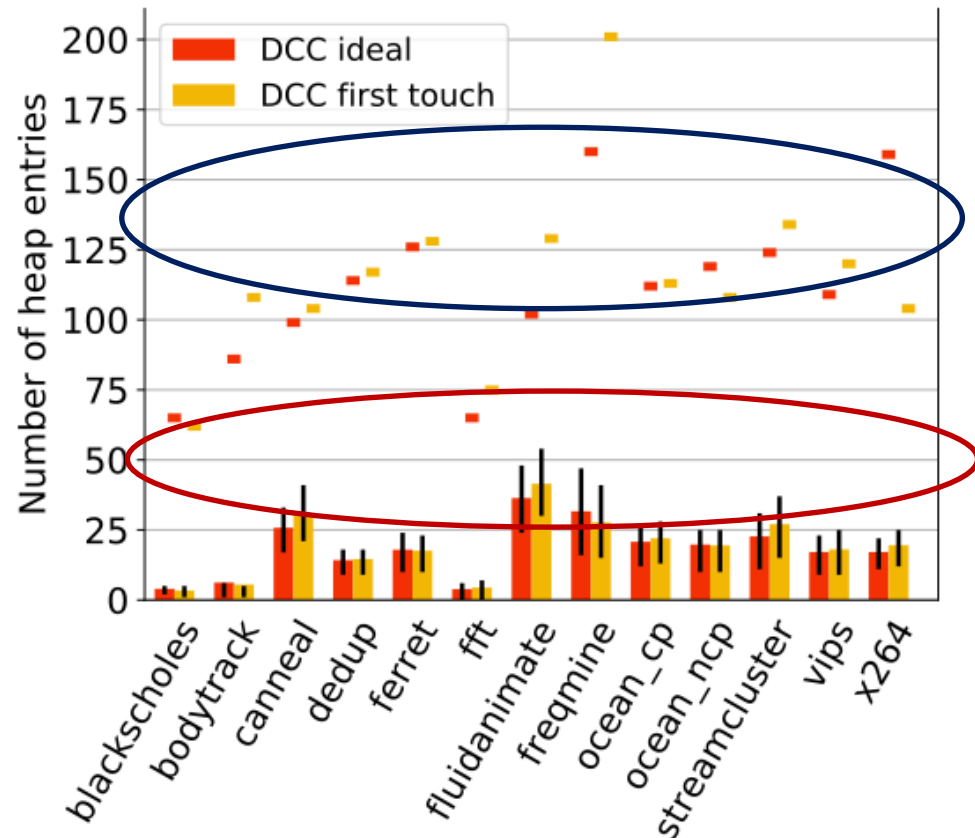
- 2 to 6 => 4

- Number of heap entries

- Simulation with an infinit heap
=> 128 entries

Maximum 100-150 entries

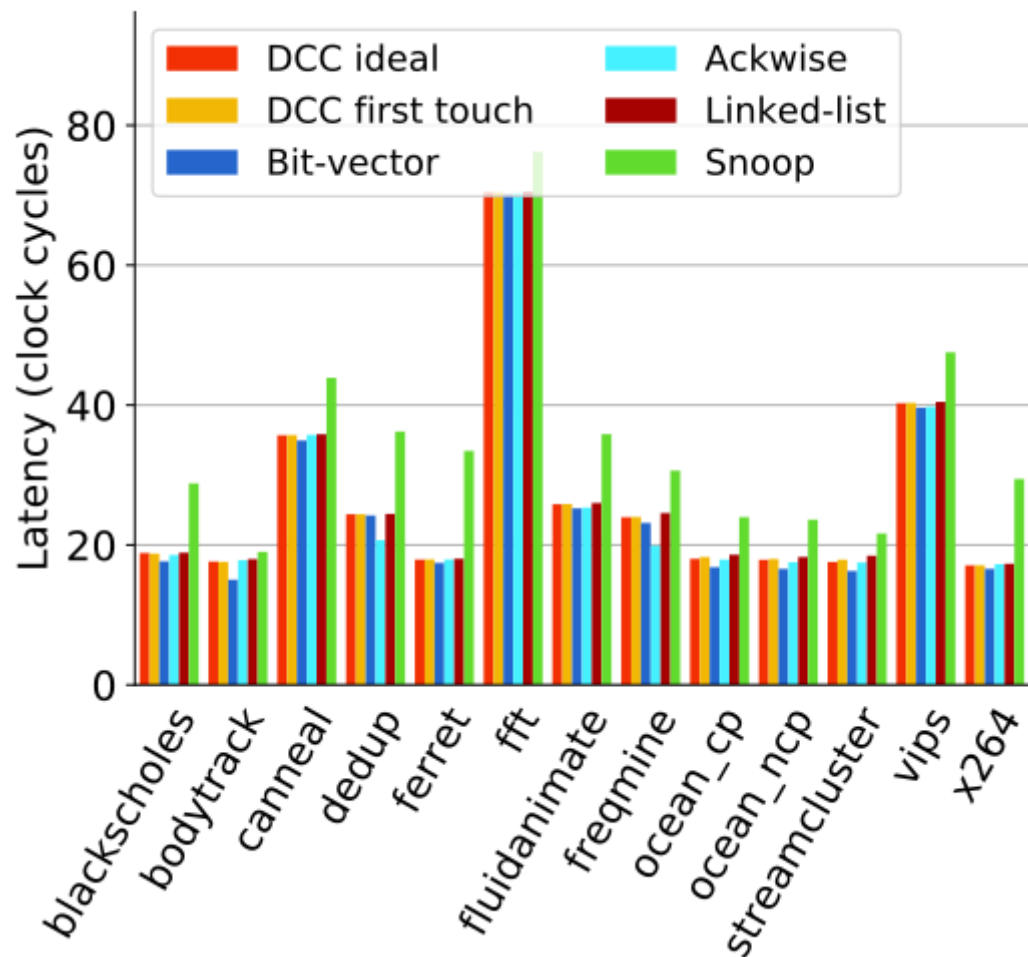
Mean < 50 entries



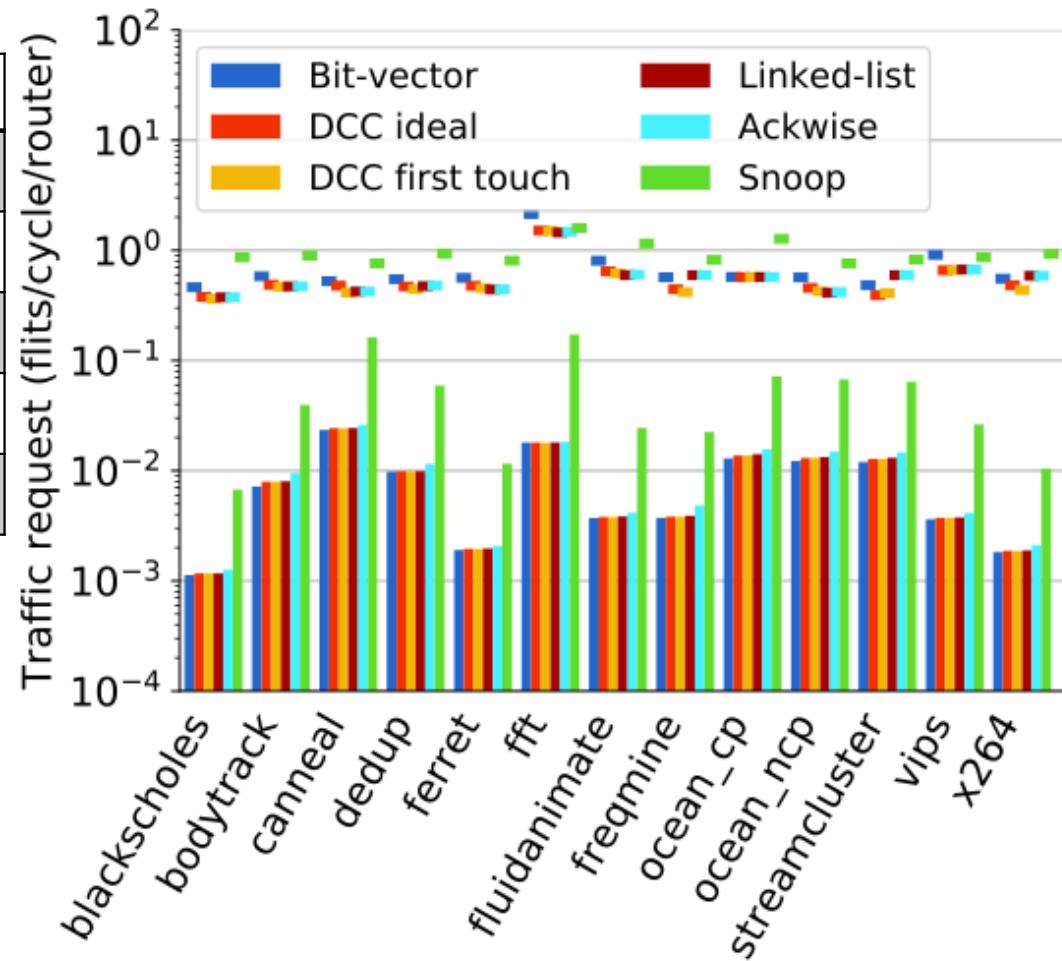
Sharing set management strategy	Directory entry size			
	Analytical formula	Number of cores		
		64	128	1024
Bit-vector	N	64	128	1024
Ackwise	$\log_2(N) \times k + 1$	31	36	51
Linked list	$\log_2(N) + \log_2(H)$	13	14	17
DCC	$\log_2(N) + C + L + \log_2(H) + 1$	33	34	37

- **DCC and Linked list**
 - Heap: $H \times (\log_2(N) + \log_2(H))$

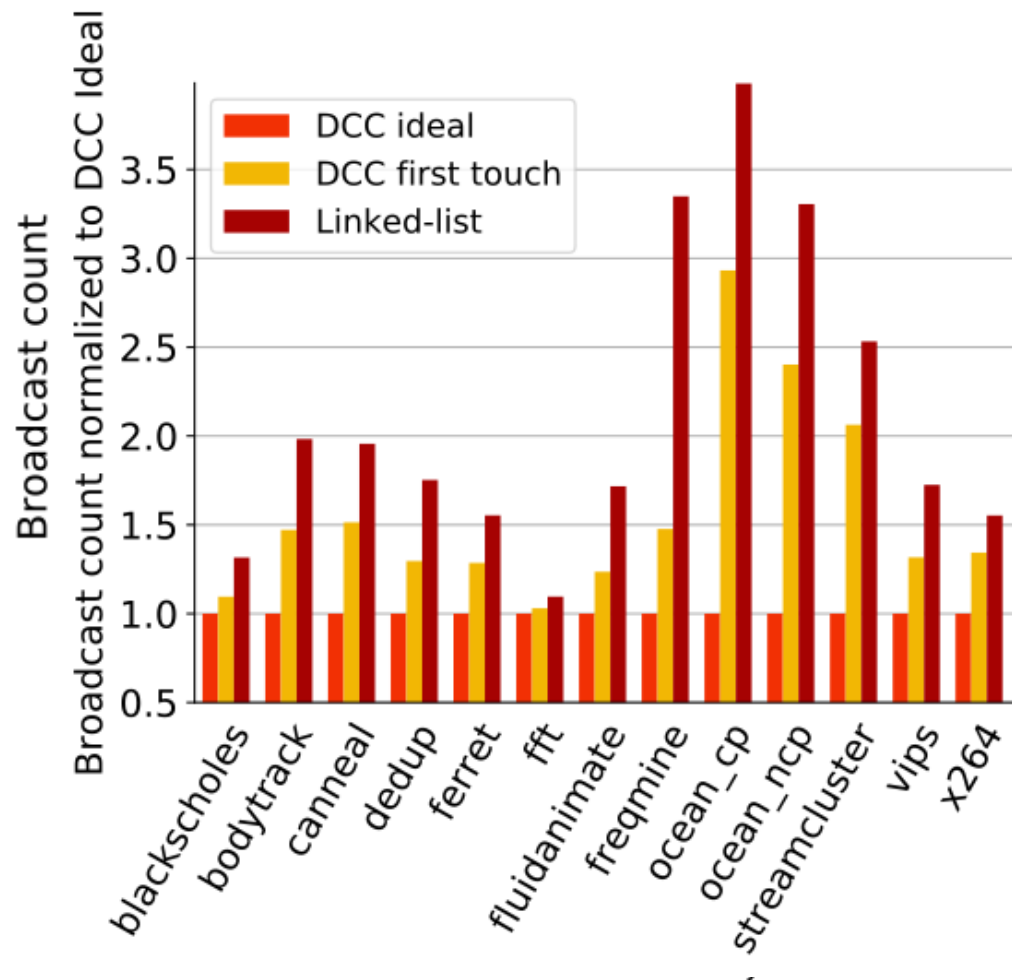
Bit-vector	0% (ref)
Ackwise	+1.9%
DCC Ideal	+4.8%
DCC 1st touch	+5.2%
Linked List	+6.4%
Snoop	+42.7%



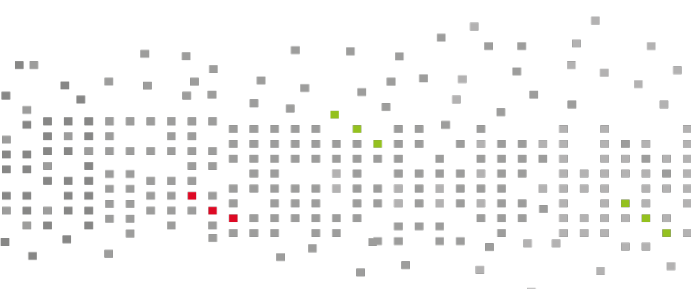
	Request	Response
Bit-vector	ref	ref
DCC Ideal	+3.9%	+4%
DCC 1st touch	+3.9%	+5.3%
Linked List	+5%	+6.5%
Ackwise	+16.8%	+16.2%
Snoop	x6.32	x40.88



DCC Ideal	0% (ref)
DCC 1st touch	+57.4%
Linked List	+114%
Ackwise	x12.67
Snoop	x789



- **DCC: compact sharing set**
 - Encode sharer in a rectangle with a bit-vector
 - Shape and rectangle origin can move during execution
 - 2 covering algorithms
- **Results**
 - Decrease hardware cost of the sharing set
 - DCC latency and traffic comparable to Ackwise
 - DCC generates less broadcast than other sharing set
 - 1 order of magnitude less than Ackwise
 - 2x less than Linked list
- **Future work**
 - New covering algorithm between ideal and 1st touch
 - Reduce directory entry size for private blocks



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Fig: Read/Read exclusive requests distribution

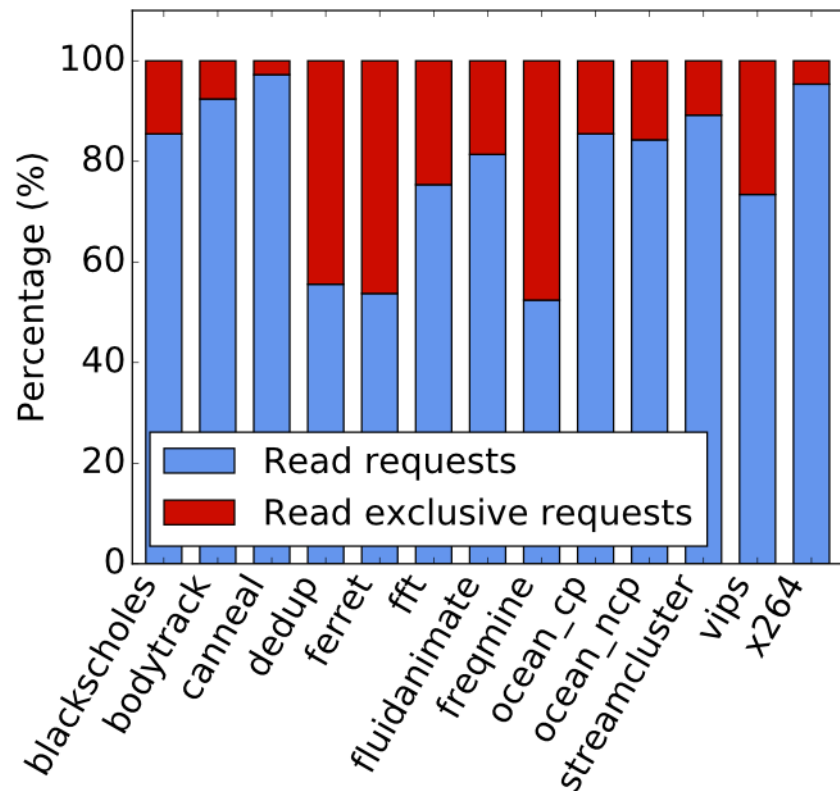
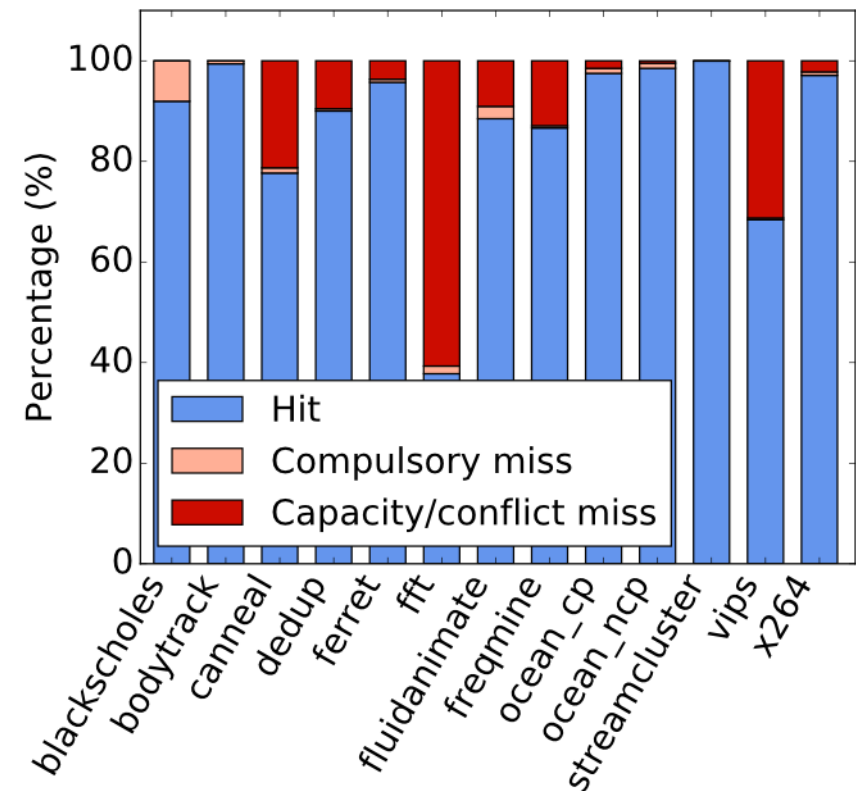
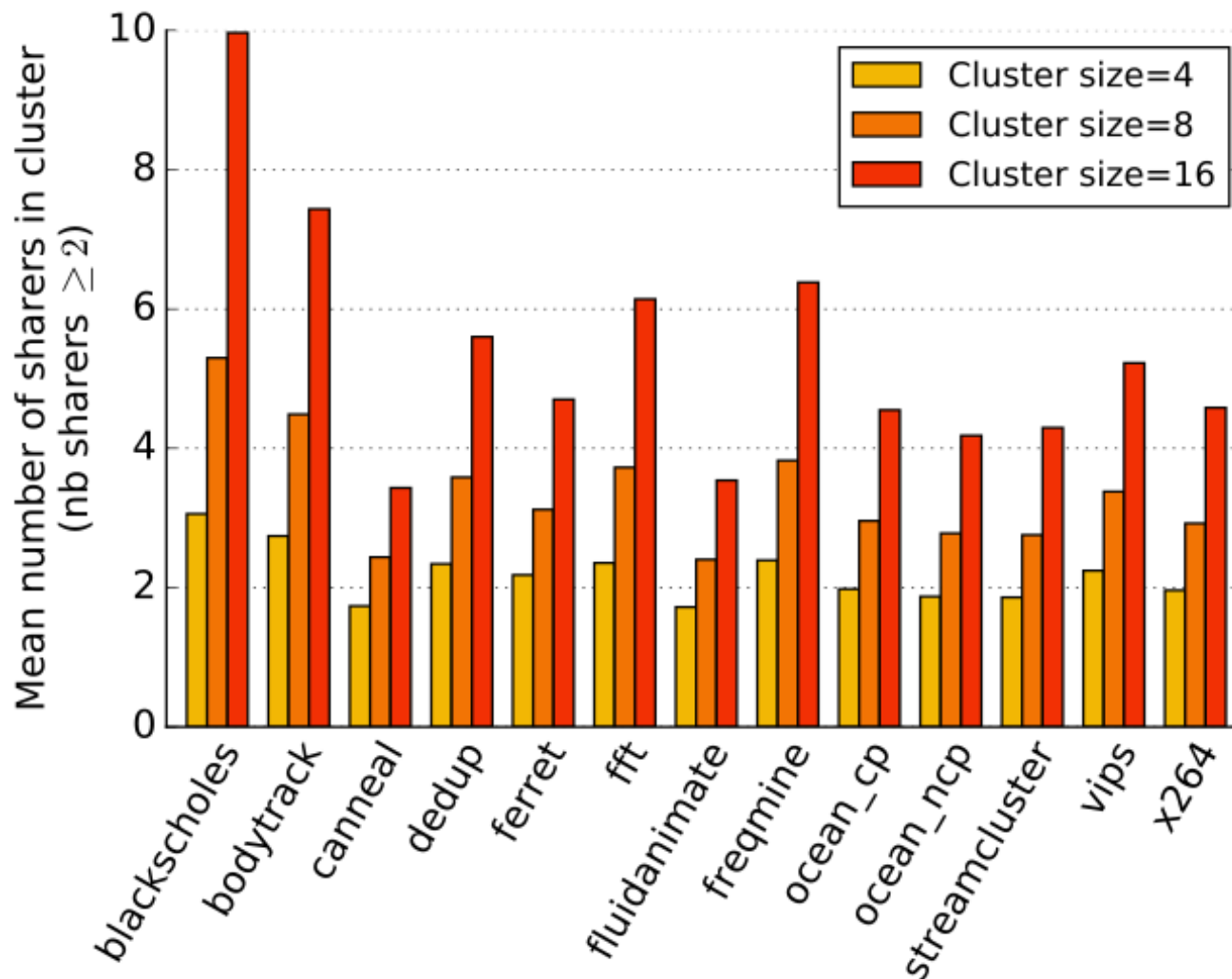


Fig: L2 hit/miss distribution



CLUSTER SIZE



LINKED LIST THRESHOLD

